

Artifact Appendix

AEGIS: Attention-Embedding Gradient Isolation Shield
Triple-Channel Gradient Masking for Privacy-Preserving Federated LLM Fine-Tuning

Paper ID: 3053

1 Artifact Appendix

Badge. We request *Artifacts Reproduced*. The artifact is a scaled-down CPU micro-GPT-2 experiment with fixed local token IDs and default **AegisDefense**. It supports the directional claim that AEGIS suppresses embedding-gradient token-set leakage versus an undefended baseline; it does not reproduce full-table ROUGE. No model, tokenizer, or dataset is downloaded at runtime.

Requirements. Docker (x86-64/arm64) or Python 3.11 on Linux/macOS. Setup 5–20 minutes; kick-the-tires <2 minutes; scaled-reproduction ~5 minutes (up to ~10 on slower hosts).

Claims. **R1:** undefended embed-norm content-token recall above threshold. **R2:** AEGIS suppresses protected recall below threshold; channels/utility remain finite.

Workflow.

```
docker build -t aegis-artifact:ndss27 .
docker create --name aegis-ktt aegis-artifact:ndss27
docker start --attach aegis-ktt
docker cp aegis-ktt:/opt/aegis/artifact-results/kick-the-tires.json .
docker rm aegis-ktt

docker create --name aegis-repro aegis-artifact:ndss27 \
  /opt/aegis/scripts/run-scaled-reproduction.sh
docker start --attach aegis-repro
docker cp aegis-repro:/opt/aegis/artifact-results/scaled-reproduction.json .
docker rm aegis-repro
```

Locally: `scripts/install-artifact.sh`, then `aegis artifact` (or `quick-test / scaled-reproduction`). Expect exit status 0 and a **Verified** line with `device=cpu` and schema version 3. Full detail is in `README.md`.